

A simulation model for container-terminal operation analysis using an object-oriented approach

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Abstract

This paper proposes a simulation model for container terminal system analysis. We assume that the container terminal consists of gate, container yard, and berth. The facilities used in the container terminal are transfer cranes, gantry cranes, trailers, and yard tractors. The simulation model is developed using an object-oriented approach, and using SIMPLE ++, object-oriented simulation software. We also consider a simple container terminal which is a reduced system of a real terminal in Pusan, Korea and we will analyze the performance of the system. © 1999 Elsevier Science B.V. All rights reserved.

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1. Introduction

Today, more than 90% of international cargo moves through seaports and 80% of seaborne cargo moves in containers through major seaports. Management of container terminal operations has thus become crucial in order to meet the demand for container traffic both effectively and efficiently. Container terminal operations can best be analyzed using queueing models. It is believed that analytical queueing models are valid only if the probability distribution of the arrival times and service times of the ships belong to the Erlang family [1]. However, the container terminal operation is difficult to check analytically with queueing models. Therefore

simulation is an effective alternative for container-terminal system analysis.

Simulation is perhaps the best tool used for any non-trivial, real world system. For analysis of complex systems, simulation is often used prior to the operation of the real world system as a mediator for a dynamic situation [2]. Therefore, simulation methodology has been recommended and chosen to analyze container terminal systems. The most popular port simulation models are the UNCTAD port model, PORTSIM, and the MIT port simulator [3]. The UNCTAD port simulation model, developed in 1969, was used to analyze port operations dealing with conventional loose cargo. PORTSIM, developed by the World Bank in the 1970's, was intended as a project appraisal tool and became useful for evaluating the costs and benefits of changing a port configuration. Developed in the

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early 1980's, the MIT port simulator is a refinement of the earlier models, as it permits the analysis of a multipurpose port entailing break-bulk cargo, bulk cargo, refrigerated cargo, and containers. However, these models are not sufficient to analyze the operations of the dedicated container ports of the 1990's. Currently, modern container terminals are equipped with modern sophisticated container handling equipment. It is therefore necessary to construct more accurate models to analyze effectively and efficiently the operations in current container terminals.

Container handling operations at the Korean container terminals have expanded considerably. Consequently, there will be an increased need for new container terminals. Also, the expansion of existing container terminal facilities will increase. The problem under consideration is whether the existing container terminal is efficient enough to handle the large container streams or whether the system using transfer cranes and gantry cranes would be more effective. In order to investigate this further, this simulation study was initiated.

The objective of this study is to develop an object-oriented simulation model to analyze the typical container terminal system (CTS) used in Pusan, Korea. This analysis includes container handling at the terminal, container transport between equipment, and equipment control. This model will also investigate the possible operation rules at the Korean container terminal.

Our model provides estimates for container terminal performance indicators and container handling equipment performance indicators including gantry crane and transfer crane utilization and container yard occupancy rate.

We assume the CTS has gate, container yard, and berth. It also includes transfer cranes (TC), gantry cranes (GC), trailers (Tr) and yard tractors (YT). We used SIMPLE ++ simulation language which is based on object-oriented programming which enables greater reusability and can also run on parallel processors.

We begin our analysis with the description of a CTS and the operations of gate, container yard, and berth. A simulation model for the CTS is presented with an object-oriented approach. A simple CTS is considered and we test and analyze its performance using the developed model.

2. System description

We assume that the CTS consists of three subsystems: gate, container yard, and berth. Container handling equipment in this system are transfer cranes, gantry cranes, yard tractors, and trailers, see Fig. 1.

The basic tasks in the operation of a CTS consists of receiving, delivering, loading, and unloading. These operations occur simultaneously and interactively.

1. Receiving operation: transporting the export containers brought by the trailers from out of the gate, to the transfer cranes in the container yards.
2. Delivery operation: lifting the import containers by the transfer cranes, placing them on the trailers in the container yards, and moving them out of the gate.

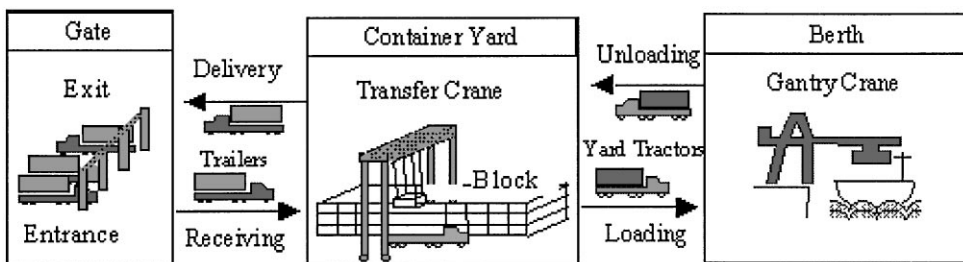


Fig. 1. The structure of a CTS.

3. Loading operation: lifting the export containers by the transfer cranes in the container yards, putting the containers on the yard tractors in the container yards and finally placing them in the ship's bay using the gantry crane.
4. Unloading operation: lifting the import containers by the gantry crane in the ship's bay and placing them on the yard tractors for the purpose of transportation to the container yards.

The management of a container terminal consists of berth allocation, yard planning, stowage planning, and logistics planning. Berth allocation controls the loading and unloading of a ship's containers. Yard planning assigns optimal allocation of storage areas for import, export and transshipment containers. Stowage planning assigns storage locations to the containers in the bay of the ship. Logistics planning assigns and coordinates the operations of the container handling equipment such as gantry cranes, transfer cranes, and yard tractors in the transportation of containers between the ship's bay and the container yard.

3. Modeling using object-oriented simulation

The dedicated tools, such as SEEWHEY, WITNESS, CADENCE, and MICROSANT deal with a restricted number of problems that exist in this industry. General-purpose simulation software and languages, such as GPSS, SIMAN, SIMSCRIPT, MODSIMII, and QNAP2 are based on a formalism which is difficult to acquire rapidly [4].

To develop the simulation model for CTS with an object-oriented approach, we used the object-oriented, simulation language SIMPLE ++ and summarized the general procedure of object-oriented simulation in this section. In an object-based world, all physical and conceptual entities are considered objects, even abstract notions, such as numbers. Each object contains a set of attributes and a set of methods. Attributes are factual descriptions of the object, while methods are procedures that enable the object to manipulate or update its attributes and communicate with other objects [5].

In addition, object-oriented models offer two other advantages over algorithmic models:

- Modular modeling: The independence of objects allows easy model transformation (i.e. object addition/deletion) without model reconfiguration.
- Generic modeling: The capability of constructing models at high levels of abstraction (i.e. transporter class instead of transfer crane) enables the model developer to initially construct generic and compact models that can be instantiated subsequently to particular specifications. Furthermore, generic units of modeling (as classes and super classes) can be aggregated to build sub-models that can be stored, enabling more complex models to be built upon request.

For modeling purposes, each component of equipment is created as a basic object (e.g. transporter object in SIMPLE ++) with properties analogous to their real-world counterparts. The use of objects in class library provides important benefits in the way of features for managing complexity. The ability to send and receive messages between objects is essential for effective communication. This creates the impression of object autonomy by only allowing objects to affect each other via message.

Class similarity between objects is expressed through a hierarchical structure. This structure allows commonality to be defined at the highest level and extended to each lower level sub-classed [6]. In this case, common attributes afford a form of modularity and extensibility by allowing lower sub-classed objects to add required specialization to all the attributes inherited from super classes.

An object-oriented environment is highly hierarchical. Each object belongs to a class, and each class to a super class [at the highest level is the abstract class(object)]. Each member of the same class shares the same attributes and methods [7]. Within a particular class, objects are differentiated by varying attribute values. A class inherits its characteristics from the parent class, but may have additional attributes and methods. Multiple inheritance is also possible in various environments.

Consequently, an object can be a member of more than one class. Inheritance is a desirable feature particularly because it reduces code rewriting.

Objects communicate with each other through message passing. Messages can be informational or declarative. Informational messages correspond to fact inquiries (e.g. obtaining the value of an object's attributes) while declarative messages reveal procedures of an event that results in updating the values of an object's attributes or in the broadcast of another message. It is important to note that there is a one to one correspondence between the messages received by an object and the set of methods within the object, thus methods can be considered subroutines, and messages as subroutine calls.

Limiting the communication between objects to only message sending and receiving enables encapsulation, also known as information hiding. For example, the methods of one object are inaccessible to another (impossible to read or write). For this reason, objects are completely independent of each other allowing the development of highly modular programs.

4. Developed simulation model

Modeling CTS using an object-oriented approach is reasonable for the following two reasons:

1. **Object and physical entity correspondence:** It is possible to establish a one to one correspondence between model objects and system entities (i.e. a transporter class with class instances corresponding to individual transporter: transfer crane, gantry crane, yard tractor, etc.). Such a correspondence between model units and physical components makes the modeling process more accessible and intuitive. This correspondence also makes the model-based reasoning process more straightforward and potentially more efficient.
2. **System hierarchy and model hierarchy correspondence:** A higher level mapping between the actual system and the model can also be achieved at the structural level. Explicit representation of that hierarchy is often important

in solving control problems. Once again, such hierarchies are easily implemented in 'class-cascading'. Hierarchical modeling also gives the additional advantage of viewing and manipulating the system model at various levels of abstraction.

4.1. System hierarchy

The system hierarchy of a CTS is described in Fig. 2.

Objects, previously defined, can be used to build hierarchies of descendant objects, with each descendant inheriting its ancestor's code and data [8]. Fig. 2 illustrates the block object code inheriting the container yard object characteristics.

4.2. Application of material flow objects

Fig. 3 describes the application of material flow objects in our CTS simulation model. The key facilities and subsystems are defined as the various types of objects. Fig. 3 includes various types of objects, which are defined below.

Movable object: it can change its position and reside in other material flow elements.

Stationary object: it cannot change its position and does not reside in other material flow elements.

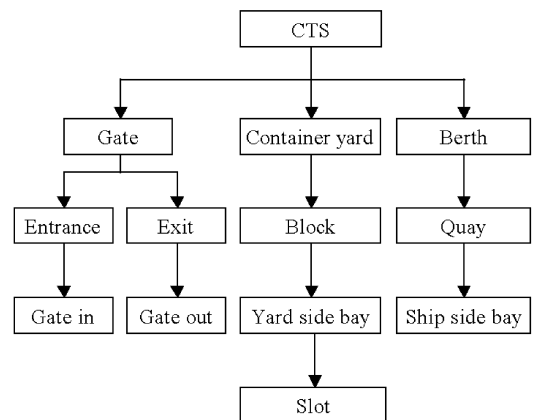


Fig. 2. System hierarchy.

All Movable & active objects have a drive component and an independent drive. Movable & passive objects do not have their own drive, therefore an entity's location can only be Movable & active objects.

Stationary & active object is a single processor that occupies a movable unit (e.g. TC, GC, and YT, etc.) and tries to pass it to the successor after processing time. Stationary & passive objects cannot move and play a role of working space.

4.3. Construction of hierarchical structure

We defined application objects as variants of the basic objects in the class library of SIMPLE ++ . Fig. 4 illustrates an example of a user defined object representing a storage element with transfer crane road, trailer roads, and yard tractor roads,

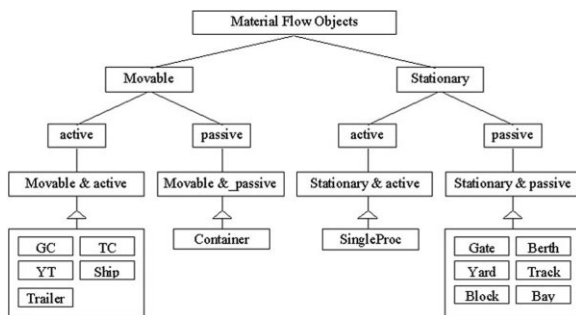


Fig. 3. Application of material flow objects.

and yard side bays. For instance, it presents a block for a container yard model. Two tracks (roads) and several stores can be employed independently to construct the new building block. In Fig. 4, Cont_list is the container list for the TC work and Bay_alloc is the allocated container list per bay. Cont_list and Bay_alloc deal with the information flow of the block in the building frame.

Using the built-in picture editor and a new icon, a corresponding animation structure can be created for the application object. It can then be used in the same way as any other basic or user defined class in the library [9].

At the next level, an entire container yard may be modeled using the new class of Fig. 4. Fig. 5 gives an example of a container yard containing road elements, TC_worklist table, and several blocks designed in Fig. 4. The road element is defined as two-direction for TC and one-direction for YT. This system will be used as an example model in the following section.

4.4. Model control using method

In order to control the system, we created control methods. Table 1 gives user defined methods to control CTS in our model. Each level's user defined methods control the interaction of objects and checks the work situation. In the highest level frame, CTS manages the container generation, the system initialization, system reset, and checks the travel roads. Methods defined in each frame

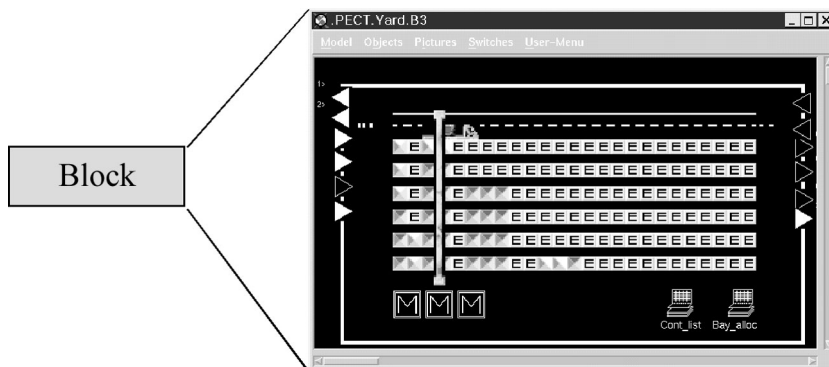


Fig. 4. Example of a user-defined object.

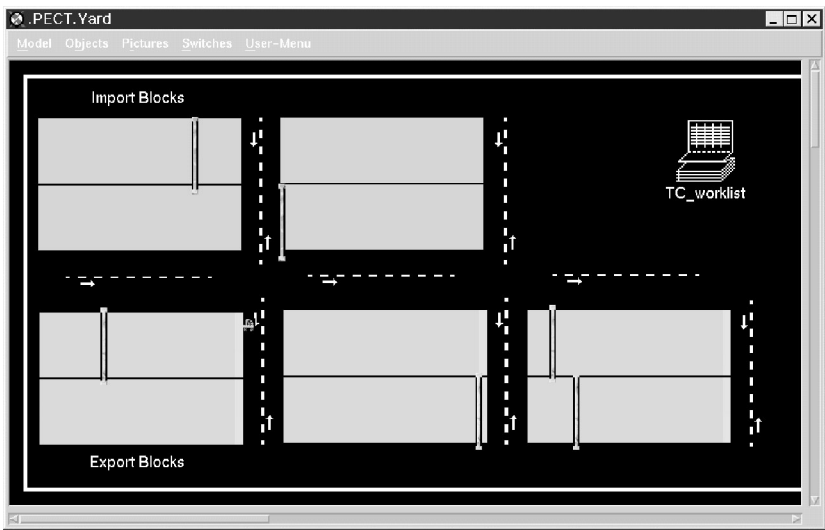


Fig. 5. A container yard with user-defined objects.

Table 1
Methods in our simulation model

Frame name	Method name	Control function
CTS	Init	System initialization
	Reset	System reset
	Produce	Generating container
	Travel	Checking the travel roads of YT and Trailer
Gate	Bay_alloc	Allocation of bay for containers
Yard	TC_control	Driving and working control of TC
	Branch2	Road control of TC, YT, and Trailer in the block
	Branch_T	Checking travel roads of YT and Trailer in the container yard
Block	Com_pos	Checking TC work position, comparing TC and YT work position
Bay	Load	Control of TC loading and unloading
Berth	GC_control	Driving and working control of GC
	Ship_gnrt	Control of ship work
Road	Road_control	Driving and destination control for TC, GC, YT, and Trailer
	Road2	Control of two-direction driving for TC and GC
	Road1	Control of one-direction driving for YT and Trailer

control each level frame. For example, Bay_alloc in Gate allocates the block and bay for the storage areas.

An example of a TC_control method for driving and working control is shown in Fig. 6. This method is defined in a block model, and it controls TC within the block. This method can control the TCs in each block.

4.5. Representation of animation

SIMPLE ++ is a simulation language which supports the animation. This section explains the animation of the simulation model. The animation of SIMPLE ++ is on-line with the simulation. We can achieve information hiding in the specification and displaying of simulation models in terms of

```

-- .Block_M.TC_control
is
local  a,b,ob1,ob2,ob3 : object;
      aa,bb,s1 : string ;  M,N,n1 : integer ;

do
ob1 := @.location;
ob2 := @.location.location;
if @.Worklist.occupied
then
  if ob2 = @.Worklist.top
  then
    @.waiting;
    if ob2.TC_Bay.pred(1).occupied
    then
      ob2.TC_Bay.pred(1).cont.continue;
    end;
  else
    a := ob2 ; aa := obj_to_str(a); M := str_to_num(aa);
    b := @.Worklist.top; bb := obj_to_str(b);
    N := str_to_num(bb); B1 := N;
    if ob2.TC_road.occupied
    then
      if aa < bb
      then
        @.move(2);
        ob3 := ob1.succ(2);
      elseif aa > bb
      then
        @.move(1);
        ob3 := ob1.succ(1);
      else
        @.waiting;
      end;
    elseif ob2.TC_road1.occupied
    then
      if aa < bb
      then
        @.move(1);
        ob3 := ob1.succ(1);
      elseif aa > bb
      then
        @.move(2);
        ob3 := ob1.succ(2);
      else
        @.waiting;
      end;
    end;
  end;
end;

```

Fig. 6. Method example for driving control of TC.

animation. Animation contributes to the validation of this system. The validation of a system consists of verifying that the action of the system (from the inputs of the system to its operation logic) is correctly represented, and has a level of accuracy and abstraction in relation to the objectives.

The main problem when viewing the simulation models is that entity paths can become complex and difficult to visualize [9]. We attempt to clarify and simplify this complexity by displaying paths for distinct entities upon a modeler's request and hiding paths inside nodes that are composed of other nodes.

The procedures to represent animation are as follows:

Step 1. The creation of an icon for class representation: these icons make an entry in the class library.

Step 2. The creation of an icon for instance representation: these icons define the detailed work state and the work type.

Step 3. The connection of subsystems: these connections define the interactions between subsystems.

Step 4. The connection of objects: these connections define the sequence that presents the relationship (predecessor/successor) between objects.

Step 5. The definition of an animation point: this definition considers the process time, travel speed, and travel distance.

Every object in the model has a personal icon. To visualize the equipment, equipment must move through the animation points. With the correct animation, visual effectiveness increases during the simulation run.

5. Experiment and analysis

We used a reduced model of PECT (Pusan East Container Terminal), which is a container terminal of the Pusan port in Korea to show an example of model building by the simulation model we developed, because the real container terminal requires massive data of terminal operation and planning. However, this model considered the values of the various parameters of facility operations and some criteria are used to evaluate the system effectiveness of the basic model.

5.1. Experiment environment

The scope of our experiment model is as follows. The gate has two entrances and one exit. The con-

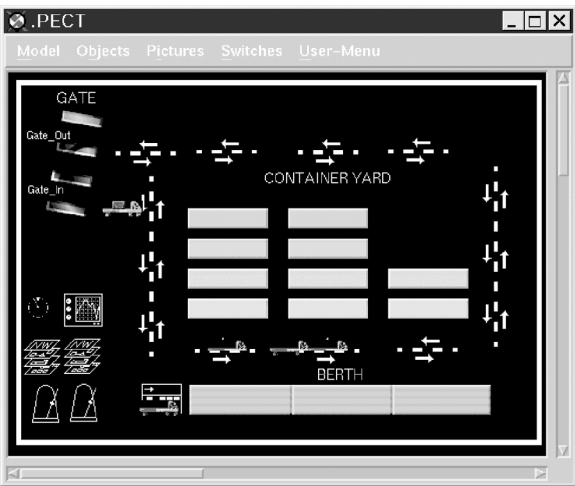


Fig. 7. Experiment model.

Table 2
Characteristics of subsystems

Gate characteristics		Yard characteristics		Berth characteristics	
Entrance	2	Number of block	10	Container handling per ship (TEU)	Export 420
Exit	1	Number of yard side bay per block	25		Import 225
Interarrival time of Trailer	Exp (2 min)			Number of ship side bay	8
Buffer capacity of Trailer	4			Interarrival time of ship	Exp (10 h)
Service time	UN (20 s, 30 s)				

tainer yard has ten blocks; consisting of four import blocks and six export blocks, as well as four transfer cranes. A block includes 25 bays; each bay consisting of 6 rows by 4 tiers. The berth has a quay and two gantry cranes. The container types are 20 and 40 foot containers (in a ratio of 45%:55%). The 20 foot container is 1 TEU (Twenty-Foot-Equivalent-Unit) and the 40 foot container is 2 TEU. Therefore the maximum capacity of one block is 600 TEU. Fig. 7 shows the configuration of our experiment model.

5.2. Input data

The input data consists of gate, container yard, berth, and equipment characteristics including basic attributes of equipments. The characteristics for each subsystem, such as gate, container yard, and berth, are summarized in Table 2. Table 3 shows the basic attributes for equipments.

5.3. Experiment results

We simulated this model for one week and the total operation times of TC and GC can be divided into their waiting time, travel time, and the working time for the loading and unloading operation. YT has travel time and waiting time but we did not consider the working time of YT. We use the following criteria for evaluating the system effectiveness and for operation analysis:

TC and GC utilization index =
$$\frac{\text{travel time} + \text{working time}}{\text{travel time} + \text{waiting time} + \text{working time}}, \quad (1)$$

Table 3
Basic attributes for equipment and their operations

Basic attribute	TC	GC	YT	Trailer
Speed (km/h)	8.04	2.7	20	20
Operation time	Exp (1 min)	Exp (2 min)		
Number of equipment	6	2	8	infinite
Drive direction	two direction	two direction	one direction	one direction

Table 4
Simulation result for experiment

Experiment condition	Parameters							
	TC utilization (%)		GC util. (%)	YT util. (%)	Trailer util. (%)	Container yard occupancy rate (%)		Average ship waiting time for berth (h/min)
	Export block	Import block				Export block	Import block	
basic attributes of Table 3	46.58	35.96	60.80	53.6	65.8	44.78	28.3	1:02
TC operation time: Exp (2 min)	52.09	49.03	50.93	54.0	61.4	51.16	25.61	1:08
speed 15 km/h								
Number of YT: 10	58.09	43.66	46.72	74.8	59.8	49.8	25.23	1:14

$$\text{YT utilization index} = \frac{\text{travel time}}{\text{travel time} + \text{waiting time}}. \quad (2)$$

The two indexes above mean the average utilization rate of the equipment. The occupancy rate of the container yard indicates the level of demand for yard services. It is defined as the percentage of the total storage level by a total yard capacity:

$$\text{Container yard occupancy rate} = \frac{\text{occupied space per unit time}}{\text{total capacity}}. \quad (3)$$

The total waiting time of a ship is the sum of the following components:

- Waiting time in the harbor for a berth.
- Waiting time at the berth for the beginning of loading and unloading operations.

The average waiting time of a ship is then the total waiting time of all the ships berthed divided by the

number of ships berthed. Average waiting time for each of the above components can be computed using the same formula:

$$\text{Average ship waiting time} = \frac{\text{number of ships berthed}}{\text{total waiting time of ships}}. \quad (4)$$

Table 4 represents the simulation results of the experiment model of the reduced CTS, including the equipment utilization, the container yard occupancy rate, and the average ship waiting time for berthing. We obtained some results of equipment utilization for three sets of equipment parameters. In this model, the average occupancy rate of container yard including the export blocks is approximately 50%. This average occupancy rate is quite variable. In most cases, the average occupancy rate is lower than 60%, while the peak occupancy rate is higher than 80%. From this result, to handle the temporary storage container at the block, the peak occupancy rate has to be considered for designing

the storage space (the number of blocks). Even if the average occupancy rate of a block is lower than 60%, the peak occupancy rate may stimulate the excess of containers over block capacity. The utilization of TC, GC, and YT is an interactive factor because the number of TC and YT are allocated to each GC. The allocation number of YT per GC is four and the allocation number of TC per GC is three. Therefore, the utilization of TC and YT are dependent on each GC. From the simulation result, we observe that GC utilization is more variable than other equipment. Therefore, we consider the GC prior to the other equipment.

The average ship waiting time for berth is 1 hour and 8 min. It is the objective of all container terminal management to minimize the ship waiting time and thereby maximize the utilization of container terminal resources such as berth, container handling equipment, and personnel. Reduced ship waiting time encourages trade and improves the competitiveness of the container terminal by providing efficient and effective services at a low cost.

6. Summary

This paper proposes a simulation model of the container terminal system which is developed using an object-oriented approach and SIMPLE ++, an object-oriented simulation software. Consequently, the developed simulation model can be easily modified or extended. To prove that our simulation model is efficient and effective, we took into consideration a simple terminal which is a reduced model of real CTS (Pusan east container terminal) and analyzed the performance of the system from the result of the reduced model. From the analysis, it has been known that there are a lot of design parameters in CTS which affect the performance and if we want to obtain reasonable CTS, we can

change the design parameters (number of YT, operation time of TC, TC speed), simulate and obtain the results iteratively by our simulation model.

Presently, container handling operations at the Korean container terminal have expanded considerably. Consequently, there will be an increased need for new container terminal and extended container terminal facilities. The problem under consideration is whether the existing container terminal is efficient enough to handle the large container streams or whether a system using transfer cranes and gantry cranes would be more efficient. In order to solve this problem and obtain a good alternative, the simulation model we developed will be useful. For further study, we are developing another CTS in which straddle carriers and automated equipments are used.

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